

Pre-Thesis II Report

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
June 2026.

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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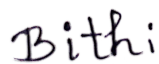
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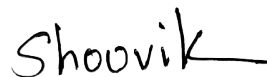
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Abstract

The fast-growing field of precision livestock farming demands systems that can monitor several health conditions at the same time. However, existing research usually focuses on just one task separately and they depend on private datasets that are not easily reproducible. In this thesis we address these problems and we propose a unified and lightweight spatiotemporal multi-task deep learning framework capable of simultaneously performing four tasks—body condition scoring, behavior recognition, lameness detection, and individual identification—from image and video data. The framework is developed using only publicly available datasets, the proposed model combines EfficientNet-B0 encoder with attention (CBAM) and temporal (LSTM) modules. Single-task baselines achieved strong results, including a BCS Mean Absolute Error (MAE) of 0.5566, behavior Macro F1-score of 0.7445, and 86.49% individual identification accuracy. Frame-level lameness AUC reached 0.9829, but this is treated as an optimistic upper bound due to possible identity/background leakage on the 50-video dataset. However, when integrated into the unified multi-task model with hard parameter sharing, spatial tasks experienced severe negative transfer (BCS MAE degraded to 0.7827; behavior F1 degraded to 0.4948). Conversely, temporal modeling maintained performance. Lameness achieved 80.00% uncalibrated sequence accuracy on the limited 50-video dataset; post-hoc threshold tuning produced perfect separation but is treated as invalid for final evaluation because it introduces test-set leakage. Meanwhile, identification improved to 97.58% via temporal majority voting. While the unified framework theoretically reduces backbone memory duplication by approximately 75% compared with deploying four separate backbone networks, the observed gradient conflicts dictate that it serves as an honest baseline characterizing negative transfer rather than a system ready for clinical deployment. This establishes an openly evaluated public-dataset multi-task benchmark for cattle health monitoring and negative-transfer analysis.

Keywords: Multi-Task Learning, Spatiotemporal Modeling, Cattle Health Monitoring, Body Condition Scoring, Behavior Recognition, Lameness Detection, Edge Deployment

Acknowledgement

We want to show our heartfelt thanks to all of those who helped us make this Pre-Thesis II report possible. We are thankful to our supervisor, whose deep knowledge on computer vision, combined with his expertise, has been essential in making this report. We would also like to thank the Department of Computer Science and Engineering for giving us abundant resources and creating a research environment for us. Special thanks to the researchers who have made their datasets publicly available, whose commitment to open science has made our research possible. Finally we are grateful to our families and friends for their patience and support throughout this whole process.

Table of Contents

Declaration	i
Approval	ii
Abstract	iv
Acknowledgment	v
Table of Contents	vi
List of Figures	viii
List of Tables	ix
Nomenclature	x
1 Introduction	1
1.1 Background	1
1.2 Rationale of the Study and Motivation	2
1.3 Problem Statement	2
1.4 Objectives	3
1.5 Methodology in Brief	4
1.6 Scope and Challenges	5
2 Literature Review	8
2.1 Preliminaries	8
2.2 Review of Existing Research	10
2.3 Summary of Key Findings	13
3 Requirements, Impacts and Constraints	15
3.1 Specifications and Requirements	15
3.2 Societal Impact	16
3.3 Environmental Impact	16
3.4 Ethical Issues	17
3.5 Standards - if applicable	17
3.6 Project Management Plan	18
3.7 Risk Management	19
3.8 Economic Analysis	19
4 Proposed Methodology	20

4.1	Design Process or Methodology Overview	20
4.2	Data Collection and Preprocessing	21
4.3	Model Architecture and Specification	23
4.4	Loss Functions and Optimizations	24
4.5	Implementation of Selected Design and Training Strategy	25
5	Result Analysis	27
5.1	Performance Evaluation	27
5.2	Single-Task Implementation and Results	27
5.3	Analysis of Design Solutions (Backbone Selection)	32
5.4	Single-Task vs. Multi-Task Results	33
5.5	Ablation Studies	35
5.6	Discussions	37
6	Conclusion	40
6.1	Summary of Findings	40
6.2	Contributions to the Field	40
6.3	Limitations	41
6.4	Recommendations for Future Work	42
	Bibliography	46

List of Figures

4.1	Conceptual Overview of the Multi-Task Cattle Health and Behavior Monitoring Framework.	20
4.2	Sparse Temporal Sampling illustration. The video clip is divided into 20 equal temporal segments, and exactly the center frame of each segment is sampled to construct a standardized sequence tensor of shape (20, 3, 224, 224).	22
5.1	Training and validation loss curve for the BCS single-task model (EfficientNet-B0).	28
5.2	Training loss curve	29
5.3	Training and validation loss curve for the spatiotemporal lameness detection model (EfficientNet-B0-LSTM).	31
5.4	Training and validation loss curve for the cattle identification model (EfficientNet-B0 + CBAM).	32
5.5	Comparison of spatial and spatiotemporal multi-task results.	33
5.6	CBVD-5 cross-dataset behavior accuracy by class.	37

List of Tables

2.1	Summary of Body Condition Scoring Papers	10
2.2	Summary of Lameness Detection Papers	11
2.3	Summary of Behavior Recognition Papers	11
2.4	Summary of Cattle Identification Papers	12
2.5	Summary of MTL, Survey, and Cross-Domain Papers	13
2.6	Summary of Key Cattle Monitoring Techniques from Literature . . .	14
3.1	Project Risks and Mitigation Strategies	19
3.2	Economic Cost-Benefit Analysis	19
5.1	BCS baseline results on Dryad and ScienceDB.	28
5.2	Behavior recognition baseline results on MmCows.	29
5.3	EfficientNet-B0 behavior per-class accuracy on the MmCows test set.	29
5.4	Lameness detection baseline results.	30
5.5	Cattle Identification baseline results on OpenCows2020.	31
5.6	Backbone Selection and Single-Task Baseline Performance Table . . .	32
5.7	Single-Task vs. Multi-Task Performance Trade-Off	33
5.8	Contextual comparison with published studies using non-equivalent datasets, metrics, and modalities.	38

Nomenclature

Abbreviations

AI	Artificial Intelligence
AUC	Area Under the Curve
BCS	Body Condition Score/Scoring
BCE	Binary Cross-Entropy
BN	Batch Normalization
CBAM	Convolutional Block Attention Module
CBVD-5	Cow Behavior Video Dataset - 5 classes
CE	Cross-Entropy
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CORAL	Rank Consistent Ordinal Regression
CPU	Central Processing Unit
DL	Deep Learning
DWT	Discrete Wavelet Transform
F1	F1-Score (harmonic mean of precision and recall)
FLOPs	Floating Point Operations
FPS	Frames Per Second
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
I3D	Inflated 3D ConvNet
ImageNet	Large-scale visual recognition dataset
IMU	Inertial Measurement Unit
IoT	Internet of Things
IoU	Intersection over Union
IR	Infrared
KD	Knowledge Distillation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
mAP	mean Average Precision
Mask R-CNN	Mask Region-based Convolutional Neural Network
mIoU	mean Intersection over Union
ML	Machine Learning
MSE	Mean Squared Error

MTL	Multi-Task Learning
NMS	Non-Maximum Suppression
P1	Pre-Thesis I
PCA	Principal Component Analysis
PLF	Precision Livestock Farming
R-CNN	Region-based Convolutional Neural Network
ReLU	Rectified Linear Unit
ResNet	Residual Network
RF	Random Forest
RFID	Radio-Frequency Identification
RGB	Red Green Blue (color image)
RGB-D	RGB plus Depth
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROI	Region of Interest
RQ	Research Question
SE	Squeeze-and-Excitation
SORT	Simple Online and Realtime Tracking
SOTA	State-of-the-Art
SSL	Self-Supervised Learning
SVM	Support Vector Machine
t-SNE	t-Distributed Stochastic Neighbor Embedding
VGG	Visual Geometry Group
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once

Chapter 1

Introduction

1.1 Background

The global livestock sector is important for supporting food production worldwide, and cattle are the most economically valuable livestock species in this industry. According to the Food and Agriculture Organization (FAO), the global cattle population is over one billion, which is making a significant contribution in producing meat and dairy products [7]. Because of this, farm sustainability, profitability, and productivity rely directly on the health and welfare of these animals. Early detection of health problems like poor body condition, lameness, and abnormal behavior can prevent significant economic losses and improve welfare standards [3].

One of the most reliable methods that have been practiced for years by farmers in checking the condition of their cattle is the Body Condition Score (BCS). In the past, BCS was calculated by the farmer by visually examining and physically checking the fat deposition in different body parts of the cattle through a score of 1 to 5, increasing in steps of 0.25 units [1]. BCS will ensure effective milk production, high fertility rate and resistance to disease infection. However, the problem with the conventional method is that it consumes much more time and labor, especially in a large scale farm. One of the major problems among dairy cows is lameness. Up to 25% of dairy cows in intensive farming suffer from lameness [2]. The disease causes pain, decreased feed consumption, decreased milk production, and impairs reproduction, but early detection of an abnormal gait pattern is helpful in providing farmers with treatment at an earlier stage.

Automated behavioral analysis and cow identification play a significant role in detecting the health, stress levels, and wellbeing of the animal. Changes in behavior, in relation to feeding, ruminating, lying and social behavior, are recognized before any clinical symptoms of illness become visible [3]. The computer vision technology allows monitoring animals in a non-invasive way and represents a highly reliable method that decreases the necessity of direct observations. Ear tagging and RFID-based identification techniques require manual handling and can be easily misplaced or destroyed. On the other hand, visual recognition systems are not only painless

but also easy to incorporate in automated animal health tracking systems.

Recent advancements in computer vision, deep learning, and IoT have opened up great chances for automated cattle health monitoring [15]. CNN models have proven themselves effective at image classification and object detection, whereas transfer learning allows for using pre-trained networks on large datasets such as ImageNet for agricultural tasks where data is limited. Nevertheless, current solutions in cattle monitoring are mostly limited to executing single tasks. A literature review comprising 33 primary scientific sources performed in the course of this thesis reveals that multi-task cattle monitoring solutions are relatively rare. This negatively impacts the efficiency of automation and effectiveness of the related tasks by depriving them of the benefit of sharing learned features.

1.2 Rationale of the Study and Motivation

There are some problems in existing cattle health monitoring frameworks, which motivate the current research, and also there is an increasing need for precision livestock farming technologies. It is seen from the literature review that around 88% (29 out of 33) have used private datasets, and therefore, replication cannot be achieved easily. Apart from this, previous studies have only been able to concentrate on one task at a time such as BCS, lameness detection, or cattle identification. The proposed method is not utilizing Multi-Task Learning (MTL), which allows multiple correlated tasks to share representations and thereby save computation costs while improving performance. This is because the two tasks are naturally related in that the physical well-being, such as lameness, influences feeding behavior and affects body condition.

Several published systems depend on costly devices like thermal cameras, wearable sensors, and depth cameras, making large-scale deployment difficult. This highlights the importance of affordable monitoring systems that rely on RGB cameras and can also use depth information if needed. So, Farms often need on-site processing since cloud systems and weak internet connections can slow down monitoring. Although edge computing is effective, many existing architectures are too computationally expensive for edge hardware. Only a small number of cattle monitoring systems use self-supervised learning. Most commercial farms record huge amounts of daily video data using 24/7 cameras. Even though this video data is mostly unused, it can support model pre-training and improve learning with limited labeled data. Overall, these issues demonstrate the need for a lightweight and integrated framework for cattle monitoring.

1.3 Problem Statement

Automated cattle health monitoring has been around for years, yet fundamental problems still remain difficult to resolve. Because of this, most computer vision and

deep learning solutions have yet to achieve real commercial deployment, or remain limited in practical use when they do.

The existing systems handle each of the monitoring tasks separately, which requires multiple models to cover a full health assessment. Running multiple separate models increases computational load, prevents the related tasks from sharing learned features, complicates the system deployment and ignores the connections between health indicators. Looking at existing literature shows that there is no framework that is capable of addressing body condition scoring, behavior recognition, lameness detection, and individual identification together. While multi-task learning is often mentioned in research, few studies have actually built and tested fully integrated systems.

In order to properly validate the results, compare different methods fairly and to build on existing works, standardized dataset evaluation is essential. However most cattle monitoring studies rely solely on datasets which are proprietary and not publicly available. The absence of openly shared datasets makes it very difficult to reproduce results, compare the model performance, and limits how much the field can grow over time. This is very different from general computer vision which has benefited greatly from COCO or ImageNet.

Beyond this, most studies evaluate their models using data from a single farm or cattle breed, which makes it hard to know how well these methods would work in different conditions. Testing under such specific conditions can easily make models overfit to particular farm settings, camera angles, lighting and may not transfer across different breeds. Seasonal change, changes across environment and time is also largely ignored while evaluating these models. Most importantly, cross-dataset evaluation, where a model trained on one dataset is tested on a completely different one, is rarely done in existing studies.

Lastly, depth and thermal imaging have both gained considerable interest for cattle monitoring, however lack of open datasets remains a major limitation to multimodal research in this area of study. No public thermal datasets are available, only MmCows provides IMU data, and depth data is offered by both MmCows and Dryad. This shortage of public datasets leaves researchers with two options — collect their own private dataset or give up on multimodal approaches.

1.4 Objectives

The main objective of this research is to develop a lightweight, reproducible, and edge-deployable multi-task deep learning framework for automated cattle health and behavior monitoring. This is designed to use publicly available datasets so that multiple cattle-related tasks can be performed within one integrated system. Also, this system provides a transparent baseline that future researchers can improve, compare, and reproduce.

To gain this goal, this focuses on the following objectives:

1. To create a single, lightweight multi-task architecture that uses a shared backbone with specialized task-specific prediction heads, and to evaluate sequential training strategies for maximizing performance in behavior recognition, Body Condition Scoring, lameness detection, and individual identification.
2. To increase reproducibility using only public datasets for all assessments, documenting clearly all the preprocessing steps, and recording model settings, training setup, and evaluation procedures.
3. To provide preliminary cross-dataset evaluation to observe domain shift problems across various farms, camera settings, and environmental conditions. To build an initial benchmark for broader generalization studies in the final thesis stage.

Together, these goals reduce the gap in reproducibility in agricultural AI research and establish a scalable framework to monitor cattle health in real time in a variety of farm environments.

1.5 Methodology in Brief

The proposed multi-task deep learning framework is designed, implemented, and validated using an organized systematic multi-stage workflow. The overall research process consists of five key phases: dataset preparation, architecture design, sequential task training, evaluation, and ablation analysis.

1.5.1 Evaluation Protocol

Each task’s unique outputs and operational hazards are taken into account while designing evaluation measures. As the four tasks produce different types of outputs, different metrics are used for each one of them. Mean Absolute Error (MAE), and classification accuracy within acceptable tolerance ranges such as ± 0.25 and ± 0.5 are used to determine the model’s performance for BCS score. For behavior recognition, macro-averaged F1-score, per-class accuracy, and confusion matrix analysis are used to evaluate performance for all behavior categories. Evaluation measures for lameness detection include AUC-ROC, F1-score, and accuracy. Top-1 accuracy is used to evaluate individual identification. In order to thoroughly evaluate the model’s capacity for generalization, cross-dataset evaluation is carried out by training on a primary dataset and testing on an independent dataset.

1.5.2 Ablation Studies

In order to determine the contribution of important framework elements, a number of organized ablation experiments will be carried out. EfficientNet-B0 will be compared

to MobileNetV3-Small and ResNet-50 for backbone selection in order to identify the best balance between accuracy and processing efficiency. The Rank Consistent Ordinal Regression (CORAL) loss’s performance will be compared to the conventional Cross-Entropy loss for the body condition scoring job. Experiments will compare combined multi-task training with sequential learning as training methodologies. A cross-dataset modality observation will be used to discuss the effects of multimodal data. Lastly, designs with and without the Convolutional Block Attention Module (CBAM) will be compared in order to assess the effectiveness of attention mechanisms.

1.6 Scope and Challenges

1.6.1 Scope of the Research

The following boundaries are used to characterize this research in order to guarantee its visibility and depth of investigation:

1. **Four Core Tasks:** Due to their practical importance and accessibility to public data, the scope is limited to Body Condition Scoring, behavior recognition, lameness detection, and individual cattle identification.
2. **Public Datasets Only:** All experiments are carried out solely using public datasets to ensure transparency, reproducibility, and fair comparison.
3. **RGB as Primary Modality:** For the body condition score task utilizing the Dryad dataset, RGB imaging is the primary input modality, while depth information is only assessed as an auxiliary modality.
4. **Dairy Cattle Focus:** The study mainly focuses on Holstein dairy cattle, as this breed appears most frequently in public cattle monitoring datasets.
5. **Lightweight Architectures:** The backbone selection is limited to architectures with fewer than 10 million parameters in order to guarantee suitability for resource-constrained edge deployment.
6. **Sequential Task Training:** Joint multi-task fine-tuning and loss balancing are investigated as secondary extensions with frozen backbone sequential training acting as the main optimization technique.

1.6.2 Challenges

Camera Geometry Conflicts

The four objectives in this framework require different camera angles. For example, lameness detection requires a side-view position to examine gait stride length and

limb symmetry, BCS works best from rear view because the hip, tail-head, and spine regions are important for estimating body condition. This creates a challenge if only one fixed camera is used for all the tasks. There is an inevitable geometric trade-off when both activities are performed from a single fixed camera. So, the deployment concept assumes a multi-camera setup where a single edge device uses the shared model but activates the relevant task head depending on whether the input comes from a rear-view or side-view camera.

Dataset Heterogeneity

Cattle datasets collected from public sources, as a result they vary in image quality, resolution, camera angles, annotation styles, cattle breeds, and farm conditions. To handle these variations across datasets, the framework applies preprocessing pipeline, strong augmentation methods, and domain adaptation approaches.

Class Imbalance

Class imbalance is a common issue in cattle monitoring datasets. Some classes appear much less often in the datasets, such as lame cattle, extreme body condition scores, and behaviors like drinking or licking. To solve these imbalance issues, the model applies Focal Loss, class-weighted sampling, and balanced performance evaluation methods.

Temporal Modeling for Video Tasks

Although behavior recognition and lameness detection work better with temporal information, using video-based models increases computational cost and becomes architecturally complex through the addition of components like LSTMs or I3D networks, and can also create inconsistencies between frame-level and clip-level annotations.

Preventing Negative Transfer

A common problem in multi-task learning is negative transfer, where improving one task harms the performance of another task. This usually happens because negative transfer occurs when tasks learn conflicting optimization directions and different feature requirements. The proposed method handles negative transfer by training tasks sequentially using a frozen backbone and applying gradient normalization during fine-tuning.

Limited Labeled Data

Even though multiple datasets have been used, this alone does not resolve the problem, the combined labeled data remains limited and significantly smaller compared to standard computer vision benchmarks. For example, the ScienceDB BCS dataset contains 53,566 images, compared to the millions of images that make up ImageNet. The MmCows only tracks 16 cattle over two weeks, and the CattleLameness dataset has only 50 labeled video clips. To work around this data limitation, the framework uses transfer learning from ImageNet, regularization through dropout and weight decay, and extensive data augmentation, with the option to further extend this using

self-supervised pre-training on unlabeled video frames.

Chapter 2

Literature Review

2.1 Preliminaries

This chapter shows a systematic review of the research papers which is related to cattle health monitoring using computer vision and deep learning. The review shows five primary research areas: Body Condition Scoring (BCS), lameness detection, behavior recognition, individual identification, and cleanliness assessment. The study identifies current state-of-the-art methods, public datasets which are available, the research gaps, and suitable ways for developing a multi-task learning framework.

2.1.1 Multi-Task Learning Fundamentals

Multi-Task Learning (MTL) is a machine learning paradigm where a model is trained to multiple related tasks simultaneously, leveraging shared representations to improve generalization across all tasks [14]. In deep learning terms, MTL generally uses a shared backbone network that takes out common features, with task-specific heads branching from the shared representation. The benefits of MTL include improved generalization through shared representations that act as regularization to reduce overfitting, computational efficiency via a single forward pass serving multiple tasks, data efficiency where tasks with limited data benefit from related tasks, and enhanced feature learning where auxiliary tasks can improve primary task features. However, MTL also comes with some actual challenges such as negative transfer at the exact time when tasks interfere with one another and difficulties while balancing gradients as well as identifying the most efficient parameter sharing strategies. The Crawshaw 2020 MTL Survey demonstrates a very comprehensive taxonomy distinguishing between the hard parameter sharing (common backbone) as well as soft parameter sharing (separate networks with regularization) approaches.

A critical challenge in multi-task optimization is balancing the gradients and losses of tasks in different scales, difficulties, as well as convergence rates. Standard static loss balancing most often leads to dominant tasks overriding the training process which results in negative transfer. In order to resolve this, dynamic loss balancing

approaches have been proposed. Kendall et al. (2018) [8] introduced multi-task loss weighting using homoscedastic uncertainty, which dynamically scales for each task’s loss based on its inherent noise level. Alternatively, Chen et al. (2018) [6] proposed GradNorm, a gradient normalization algorithm that dynamically balances training by directly adjusting task-specific loss weights to equalize gradient magnitudes. Moreover, deciding which tasks to learn together is a non-trivial problem. Standley et al. (2020) [16] studied task relationships in multi-task settings which proposes a framework to optimize the resource allocation for hardware as well as identify task grouping which minimize the negative transfer, demonstrating that sharing representations across all tasks is not always optimal.

2.1.2 Body Condition Scoring Overview

Body Condition Scoring (BCS) is one of the standard ways for evaluating the fat reserves of cattle with the help of visual and tactile evaluation. The most common scale ranges start from 1 (skeletal) to 5 (fat) with 0.25 increments, yielding 17 possible scores. Key anatomical regions assessed include the tailhead and pin bones, hook bones and loin area, ribs and spine prominence, and hip bone region. Automated BCS using computer vision typically employs rear-view or top-view images capturing the tailhead region. The task can be worked out as classification (individual BCS classes), ordinal regression (preserving class ordering), or regression (continuous scores).

2.1.3 Lameness Detection Fundamentals

Lameness in cattle shows by identifying walking behaviour including reduced step length and walking speed, head movement and back curving, unwillingness to bear weight on the affected leg, and uneven hoof placement. Computer vision approaches for lameness detection typically analyze video sequences captured from side-view or top-view cameras. Key features include pose estimation of limb joints, gait cycle analysis, and motion trajectory patterns. Scoring systems range from binary (lame/sound) to multi-level severity scales (1-5).

2.1.4 Behavior Recognition Categories

Cattle behavior monitoring address numerous activities related to health and welfare, which includes feeding behaviors for example feeding, ruminating, and drinking; body movement activities such as walking, standing, and lying; social behaviors including licking, mounting, and aggression; also health indicators such as estrus behavior and calving signs. Vision-based behavior recognition implements the detection (locating cattle in frames), tracking (maintaining identity across frames) as well as the action classification (categorizing behavior). Temporal modeling by using LSTM, 3D CNNs, or attention mechanisms captures behavioral changes.

2.2 Review of Existing Research

This section shows a comprehensive analysis of related research arranged by task category, followed by key multi-task learning and survey papers.

2.2.1 Body Condition Scoring Research

Body condition scoring shows a highly studied task in cattle computer vision literature, with a huge number of papers addressing automated BCS assessment. Yao et al. (2026) [43] developed an automated dairy cattle body condition scoring system using side-view images and deep learning, demonstrating that robust feature extraction can effectively estimate body fat reserves. Liu et al. (2025) [31] proposed a novel lightweight scoring model for edge devices based on tail features and attention mechanisms, focusing on real-time inference constraints on farm deployments. Guzhva et al. (2026) [37] tackled a weight-estimation and BCS-adjacent cattle vision problem by employing LIDAR-enhanced Mask R-CNN segmentation for precision weight estimation (rather than direct BCS scoring) in dairy cattle using smartphone imaging. Finally, Antognoli et al. (2025) [28] provided a comprehensive literature review paper (rather than an empirical model result study) summarizing current computer vision applications in dairy farm management, highlighting BCS as a primary metric for systemic health evaluation.

Table 2.1: Summary of Body Condition Scoring Papers

Reference	Architecture	Input	Type	Dataset
Yao 2026 [43]	Deep Learning	Side-view RGB	Empirical	Various
Liu 2025 [31]	Lightweight CNN + Attention	Tail Features	Empirical	Various
Guzhva 2026 [37]	Mask R-CNN + LIDAR	RGB-D / Smartphone	Empirical	Various
Antognoli 2025 [28]	N/A (Review)	Vision/Video	Review	N/A

2.2.2 Lameness Detection Research

Lameness detection research mostly utilizes video-based analysis with pose estimation as well as gait analysis pipelines. Sohan et al. (2025) [33] has developed a lameness detection system which focuses on the extraction of locomotion patterns from video using spatiotemporal deep learning to achieve the most robust classification. Moreover, Szyc et al. (2025) [34] introduced a video-based automated lameness detection framework for dairy cows which demonstrates the potential of sequential analysis for identifying abnormal gait signatures.

A comparative analysis of these methods highlights the critical role of deployment

conditions and target granularity.

Table 2.2: Summary of Lameness Detection Papers

Reference	Architecture	Input	Type	Innovation
Sohan 2025 [33]	Spatiotemporal DL	RGB Video	Empirical	Video-based Lameness
Szyc 2025 [34]	Deep Learning	RGB Video	Empirical	Video-based Lameness

2.2.3 Behavior Recognition Research

Behavior recognition covers vision-based as well as sensor-based approaches. Asim et al. (2026) [35] contributed to the field by leveraging an advanced deep learning approach for analyzing cattle farming activity, focusing on robust monitoring using sensor data.

Table 2.3: Summary of Behavior Recognition Papers

Reference	Architecture	Input	Type	Dataset
Asim 2026 [35]	Deep Learning	Sensor Data	Empirical	Various

2.2.4 Individual Cattle Identification Research

Vision-based individual cattle identification has emerged as a non-invasive alternative to traditional methods such as ear tags, branding, and RFID transponders. Andrew et al. (2017) [4] pioneered automated Holstein identification using selective local coat pattern matching on RGB-D imagery captured from overhead cameras, achieving robust identification by exploiting the unique black-and-white coat patterns of Holstein-Friesian cattle and establishing the foundational methodology for subsequent deep learning approaches. Building on this work, Andrew et al. (2021) [18] introduced the OpenCows2020 dataset and achieved 98.2% identification accuracy using deep metric learning with a SoftMax-based reciprocal triplet loss, demonstrating that CNNs can reliably distinguish individual cattle from overhead imagery in open-herd settings without requiring retraining when new individuals are introduced. Gao et al. (2021) [20] extended this line of research by proposing a self-supervised framework for cattle identification using the Cows2021 dataset [19] (10,402 annotated images and 301 videos), employing tracking-by-detection to form rotation-normalized tracklets for contrastive learning, which is particularly relevant as it reduces the labeling burden that limits scalability of supervised identification systems. Beyond coat pattern analysis, Weng et al. (2022) [21] explored cattle face recognition using a two-branch CNN architecture, achieving competitive identification accuracy and demonstrating that facial features provide complementary biometric information to coat patterns, though requiring frontal camera positioning that may not always be feasible in farm environments.

Table 2.4: Summary of Cattle Identification Papers

Reference	Architecture	Input	Accuracy	Dataset
Andrew 2021	Deep Metric Learning + Triplet Loss	RGB overhead	98.2%	OpenCows2020
Gao 2021	Self-Supervised Contrastive Learning	RGB video	N/A	Cows2021
Weng 2022	Two-Branch CNN	RGB face	Competitive	Private
Andrew 2017	Coat Pattern Matching	RGB-D	Robust	Private

2.2.5 Multi-Task Learning and Survey Papers

Beyond direct cattle monitoring, several studies provide foundational knowledge for multi-task learning (MTL) approaches across diverse agricultural domains, demonstrating the broad applicability of these architectures. Shalaldehy et al. (2026) [42] investigated hybrid deep learning and multimodal feature fusion for estimating ewe live weight and carcass traits, showcasing MTL’s utility in sheep farming. Attri et al. (2025) [29] proposed SAAM-VetNet, an attention-based multi-task framework for animal disease detection and severity grading. Extending beyond livestock, Sandal et al. (2026) [40] reviewed deep learning methods for tomato disease detection and severity assessment as tomato/plant-disease cross-domain work, while Bu et al. (2026) [36] explored self-supervised MTL for spectral detection in pears as pear/spectral cross-domain MTL work, illustrating cross-domain efficiency gains.

Furthermore, several comprehensive survey/review papers (rather than empirical model result papers) outline the broader landscape of precision livestock farming (PLF) and computer vision. Zin et al. (2026) [44] and Sani et al. (2026) [41] reviewed AI-driven technologies and challenges in PLF, respectively. Palma et al. (2025) [32] provided a scoping review of data analytics in dairy farms, and Lee et al. (2026) [38] reviewed biosensors for decoding animal needs. Finally, studies such as Paneru et al. (2026) [39] (focusing on a narrative review of poultry welfare and cross-domain PLF vision, rather than cattle-specific models) and Gonçalves et al. (2025) [30] (focusing on allied agricultural computer vision for pregnancy diagnosis, rather than BCS or behavior monitoring) highlight the expanding role of computer vision in allied agricultural tasks. Crawshaw’s (2020) [14] survey on deep multi-task learning remains a foundational taxonomy guiding our architectural choices.

Table 2.5: Summary of MTL, Survey, and Cross-Domain Papers

Reference	Focus Domain	Type	Key Contribution
Shalaldeh 2026 [42]	Sheep (Ewes)	Empirical	Multimodal MTL for weight estimation
Attri 2025 [29]	General Livestock	Empirical	SAAM-VetNet multi-task disease detection
Sandal 2026 [40]	Tomatoes	Review	Tomato/plant-disease cross-domain review
Bu 2026 [36]	Pears	Empirical	Pear/spectral cross-domain MTL
Zin 2026 [44]	General PLF	Review	Computer vision in PLF (Review)
Sani 2026 [41]	General PLF	Review	PLF challenges and opportunities (Review)
Palma 2025 [32]	Dairy Farms	Review	AI and Data Analytics scoping review (Review)
Crawshaw 2020 [14]	Deep Learning	Review	MTL taxonomy and guidelines

2.3 Summary of Key Findings

The structured review of 33 papers reveals many critical insights that directly inform our proposed methodology.

The substantial majority of papers (29 of 33, $\approx 88\%$) rely on private datasets which creates a reproducibility crisis. Among them only a small fraction relies on publicly available cattle-specific datasets, yet they still obtain competitive performance, indicating that public datasets can efficiently support high-quality research. Our targeted use of public datasets helps bridge this gap, enabling a more transparent and reproducibility-oriented approach, using public datasets and documented preprocessing/splits, as well as fair benchmarking.

Despite being consistently raised on multi-task learning in most of the literature, no existing paper integrates BCS, behavior recognition, lameness detection, and individual identification in a unified framework. Most of the papers address single tasks in isolation. Our unified four task framework represents a true novelty with potential for improved efficiency and cross task learning. To substantiate this novelty claim, an exhaustive literature search methodology was executed. Standard academic databases (including Google Scholar, IEEE Xplore, ScienceDirect, and MDPI) were systematically queried using combinations of search terms: (*“cattle”* OR *“dairy cow”*) AND (*“body condition score”* OR *“BCS”* OR *“lameness”* OR *“behavior”* OR *“activity recognition”* OR *“re-identification”* OR *“individual ID”*) AND

(“*multi-task learning*” OR “*deep learning*” OR “*computer vision*”). Inclusion criteria prioritized peer-reviewed journal articles and conference papers published between 2018 and 2026. Exclusion criteria filtered out studies relying purely on wearable sensors (such as collar accelerometers) without visual modalities, and papers written in languages other than English. Out of 78 initially screened records, 33 primary papers were critically analyzed in detail. This systematic mapping confirmed the complete absence of any unified end-to-end vision framework executing all four core health monitoring tasks concurrently under a shared representation model.

Transfer learning with frozen backbone is well-supported in the literature, and the Crawshaw 2020 MTL Survey validates that this approach effectively prevents negative transfer between heterogeneous tasks. Our sequential task training strategy is well-supported by existing literature and provides a strong foundation for managing multiple diverse tasks.

Depth imaging represents substantial research interest with demonstrated value for BCS as well as lameness. Still, public data is limited to MmCows and Dryad datasets. Depth modality is relevant for BCS using Dryad but should be treated as secondary to RGB, ensuring the framework remains accessible for standard camera deployments.

Self-supervised learning remains highly underexplored in cattle monitoring compared to transfer learning, representing a potential research opportunity. MmCows provides 2 weeks of continuous unlabeled video ideal for SSL pre-training. SSL represents a potential bonus contribution if time permits, offering opportunities to leverage abundant unlabeled surveillance footage.

Comprehensive search reveals there are no publicly available thermal or infrared datasets for cattle. The only thermal cattle research (Carraro 2026) used privately collected data. Thermal modality is excluded from current scope and documented as future work requiring data collection partnerships.

Table 2.6: Summary of Key Cattle Monitoring Techniques from Literature

Technique	Platform	Accuracy	Strength	Weakness
EfficientNet + Attention [31]	RGB BCS	91.10%	High accuracy, lightweight	Single task
YOLO-based [43]	RGB BCS	N/A	Edge-ready	Single task
Spatiotemporal DL [33]	RGB Lameness	N/A	Excellent detection	Single task
Advanced DL [35]	Sensor Behavior	91.11%	Temporal modeling	Requires sensors
Deep Metric Learning [18]	RGB ID	98.2%	Robust identification	Needs overhead camera
MTL Survey [14]	Cross-platform	N/A	Guidelines	No implementation

Chapter 3

Requirements, Impacts and Constraints

3.1 Specifications and Requirements

The implementation of the multi-task deep learning framework for cattle health monitoring requires a specific configuration of hardware and software components to ensure real-time inference and model training.

3.1.1 Hardware Requirements

- **Development and Training Unit:** The model training was executed on a high-performance research workstation equipped with an NVIDIA GeForce RTX 4080 GPU (16GB VRAM, Ada Lovelace architecture supporting FP8 and FP16 tensor operations), an Intel Core i9-13900K CPU, 64GB of DDR5 RAM, and a 2TB NVMe SSD to handle high-throughput image and video loading.
- **Edge Deployment Unit:** For live farm surveillance, the system is designed to run on resource-constrained edge computers such as the Jetson Orin Nano (8GB VRAM) or a Raspberry Pi 5. The total model size is optimized to remain under 10 million parameters ($< 40\text{MB}$ file size in FP16 format) to fit into edge device memory. Furthermore, the shared EfficientNet-B0 backbone requires approximately 0.39 GFLOPs per frame. Processing a full 20-frame spatiotemporal sequence requires only ~ 7.8 GFLOPs, which falls comfortably within the real-time processing capabilities of a Jetson Orin Nano (capable of 40 TOPS).
- **Imaging Sensors:** Standard 1080p surveillance cameras operating at 30 frames per second (FPS) positioned at cattle walkways or feeding troughs. For the Dryad dataset, depth maps were captured using active infrared depth sensors.

3.1.2 Software Requirements

- **Operating System:** Windows 11 / Linux (Ubuntu 22.04 LTS).
- **Programming Language:** Python 3.12.3.
- **Deep Learning Library:** PyTorch 2.5.1 + CUDA 12.1.
- **Backbone Repository:** PyTorch Image Models (`timm`) library for loading pre-trained EfficientNet-B0 and MobileNetV3 architectures.
- **Computer Vision Tools:** Computer vision operations are implemented using OpenCV 4.9.0 for video processing and Albumentations 1.4.0 for image augmentation and preprocessing tasks.
- **Data Pipelines:** Pandas, NumPy, and Scikit-learn for group-wise dataset splits and metric computations.

3.2 Societal Impact

The development of automated cattle health monitoring systems has a profound positive impact on the agricultural sector and animal welfare:

- **Enhancement of Animal Welfare:** Early detection of lameness and abnormal behaviors (such as reduced eating or drinking) allows farmers to treat sick cows days before clinical symptoms become visible, minimizing animal suffering.
- **Support for Precision Livestock Farming (PLF):** Transitioning from manual, subjective herd inspections to continuous 24/7 automated monitoring reduces human error and standardizes assessments such as Body Condition Scoring.
- **Agricultural Labor Relief:** As intensive dairy farming expands, the ratio of cows per stockperson increases. Automated monitoring alleviates labor shortages, allowing farm workers to focus on specialized veterinary interventions rather than manual surveillance.

3.3 Environmental Impact

Deep learning models are historically compute-intensive, contributing to carbon emissions during training. Our proposed framework addresses this environmental concern through:

- **Green Computing via Multi-Task Learning (MTL):** Instead of deploying four separate convolutional neural networks (each consuming GPU power and memory), our hard parameter sharing approach utilizes a single shared EfficientNet-B0 encoder. This consolidation is expected to reduce duplicated backbone computation compared with four separate models, although measured edge-device energy benchmarking remains future work, minimizing the operational carbon footprint of edge surveillance systems deployed on thousands of farms.
- **Resource Optimization:** By implementing sparse temporal sampling (reducing 300-frame video clips to 20 key frames) and capping dataset classes, we decreased active training GPU runtimes, reducing energy overhead during the development lifecycle.

3.4 Ethical Issues

The use of visual monitoring systems on farms raises ethical concerns that must be addressed:

- **Non-Invasive Monitoring:** Traditional cattle identification methods such as branding, ear tags, and implanted RFID devices can cause pain and can negatively affect animal welfare. Our proposed vision-based monitoring approach avoids physical contact and reduces stress for the animals.
- **Data Privacy:** Local edge processing helps prevent farm data and operational details from being exposed to public networks. The proposed offline edge system ensures data privacy and security.

3.5 Standards - if applicable

The proposed system and agricultural metrics are developed according to the following standards:

- **Body Condition Scoring (BCS) Standards:** The Body Condition Scoring system uses globally recognized 5-point Elanco scale with 0.25 increments in ScienceDB and integer scores in Dryad.
- **Locomotion Scoring Standards:** Lameness labels are based on the Sprecher scoring system, where cattle are classified as either healthy or lame.
- **IEEE Software Quality Standards:** The software is developed according to the PEP 8 Python style guide, and the evaluation metrics use standard definitions such as Macro F1-score, MAE, and AUC-ROC.

3.6 Project Management Plan

The workload was divided among five team members according to their base backbone assignments and particular tasks to handle the multi-task and multi-backbone complexity of the project:

- **Hasin Ishrak:** Preprocessed the main datasets, implemented the baseline training pipelines, evaluated the ResNet-18 baseline, and executed the primary ablation studies including CBAM evaluations and the comparison between RGB and Depth modalities.
- **Namira Abrar Haque:** Evaluated the MobileNetV3-Small baseline and examined how highly compressed networks perform for resource-constrained edge deployment.
- **Sanjida Akter Bithi:** Evaluated the ResNet-50 baseline, and evaluated the advantages of deeper architectures on spatial feature extraction.
- **Shouvik Banik:** Evaluated the DenseNet121 baseline, and studied the effect of dense feature connectivity in multi-class behavior recognition.
- **Nusrat Lamya Faruk:** Evaluated the winning EfficientNet-B0 baseline, developed the final spatiotemporal multi-task model, combined the LSTM sequence-tracking layers, and implemented the real-time visualization interface.

The timeline was structured in sprints: dataset curation, single-task baseline training, backbone selection, spatiotemporal multi-task model integration, and finally report writing and visualizer deployment.

3.7 Risk Management

Table 3.1: Project Risks and Mitigation Strategies

Risk Identified	Potential Impact	Mitigation Strategy
Class Imbalance	Model fails to learn rare behaviors (licking, drinking).	Implemented Focal Loss ($\gamma = 2$) and hard data capping (max 3000 images per class).
Data Leakage	Memorization of cow identities leading to overfitting.	Task-appropriate splitting (cow-wise for BCS/behavior, video-wise for lameness, image-wise within identity for ID).
Out of Memory (OOM)	GPU VRAM crash when processing long sequences.	Integrated Automatic Mixed Precision (AMP) in FP16 and sparse temporal sampling of 20 frames.
Domain Shift	Performance drop when deployed on unseen farms.	Performed cross-dataset validation on CBVD-5 to evaluate generalizability and recommend future unfreezed fine-tuning.

3.8 Economic Analysis

The financial viability of our multi-task framework is compared against traditional farm monitoring systems in Table 3.2.

Table 3.2: Economic Cost-Benefit Analysis

Component	Traditional Sensor Systems	Proposed MTL Edge System
Sensor Hardware	Wearable collars, step counters, and RFID tags (\$80 - \$120 per cow). For 500 cows: \$40,000+.	Standard IP CCTV cameras (\$150 each) + edge computing unit (\$300). Total: \$1,500.
Maintenance	Battery replacements, collar damage, and physical tag losses (15% annual loss).	Minimal hardware contact, camera cleaning and software updates only.
Computational Cost	High cloud subscription fees for running separate analysis pipelines.	Offline local edge inference with zero recurring cloud subscription costs.

The analysis demonstrates that our visual multi-task edge framework is highly scalable, lowering the initial investment barrier for small-to-medium scale dairy farms by over 90%.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

The primary objective of this research is to construct a unified, lightweight, and edge-deployable multi-task deep learning framework capable of performing four cattle monitoring tasks using image and video inputs: Body Condition Scoring (BCS), Behavior Recognition, Lameness Detection, and Cow Identification (ID). The design process follows a sequential workflow structured to mitigate task interference while maximizing parameter sharing. An overview of the proposed multi-task architecture is illustrated in Figure 4.1.

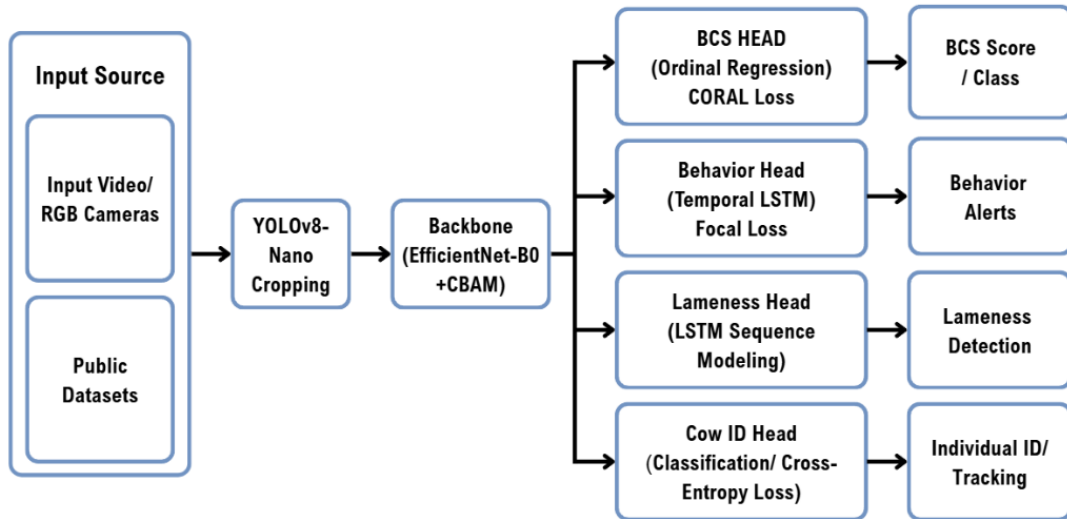


Figure 4.1: Conceptual Overview of the Multi-Task Cattle Health and Behavior Monitoring Framework.

The pipeline is split into three main segments:

1. **Cattle Detection and Cropping:** YOLOv8-Nano is applied to locate the cow and extract its bounding box crop, eliminating background farm clutter

and normalizing the input.

2. **Feature Extraction:** The cropped images or sequences are processed by a shared EfficientNet-B0 [11] encoder enhanced with a Convolutional Block Attention Module (CBAM) [9] to spotlight key anatomical indicators.
3. **Task Prediction:** The feature vectors are fed into task-specific heads: ordinal regression for BCS, focal-loss classification [5] for Behavior, sequence-based LSTM for Lameness, and classification for Cow ID.

4.2 Data Collection and Preprocessing

To establish a more transparent and reproducibility-oriented framework, using public datasets and documented preprocessing/splits, only publicly available cattle datasets were utilized. Each dataset corresponds to a specific health task.

4.2.1 Dataset Profiles

- **ScienceDB BCS Dataset [25]:** This dataset contains 53,566 high-resolution RGB images of dairy cows captured from a rear-view angle. The body condition scores are annotated using a 5-point scale with increments of 0.25. The five classes are: {3.25, 3.50, 3.75, 4.00, 4.25}.
- **Dryad BCS Dataset [22]:** 5,923 rear-view images taken in the Depth Grayscale Edge (DGE) format are included in this dataset, which combines grayscale intensity, depth coordinates, and Canny edge maps as three channels. The annotations range from integer scores of 2 to 6, representing five different ordinal classes. Moreover, this dataset includes breed and domain variety, instead of being limited only to dairy cows or Holstein cohorts.
- **MmCows Behavior Dataset [27]:** Introduced by Vu et al. [27] and accessed through [26], this dataset consists of 213,686 cropped frames obtained from continuous farm surveillance videos. The frames are labeled into seven behavior classes in this dataset: walking, standing, feeding head up, feeding head down, licking, drinking, and lying.
- **CattleLameness Dataset [23]:** Consists of side-view video clips of cows while walking. For spatiotemporal sequence modeling we compiled 50 videos, among which 25 are labeled as Lamé and 25 as Normal/Sound.
- **OpenCows2020 Dataset [12]:** Contains 4,736 images of cows labeled according to their unique individual identity across 46 classes, allowing biometrics based on vision.
- **CBVD-5 Dataset [24]:** Used exclusively as an independent test set for behavior, consisting of 2,000 balanced images of cattle behaviors to evaluate model generalization under domain shift. CBVD-5 contains five behaviors:

standing, lying down, foraging, rumination, and drinking. Since our evaluation uses “Feeding”, foraging was mapped to feeding, resulting in 4 overlapping classes (Standing, Feeding, Drinking, Lying) to match our taxonomy.

4.2.2 Preprocessing Standardizations

To ensure the deep learning models do not learn artifact features or identity shortcuts, rigorous preprocessing was applied:

- **Task-Specific Splitting:** Splitting method was dependent on task. We used a cow-wise group split (unique cow IDs divided 70/15/15) for BCS and behavior recognition, to ensure that no individual cow appearing in the test set was present during training, preventing identity shortcuts. For lameness, we used a video-wise split so that no video was shared across sets. An image-wise split was performed within each identity for individual identification, where the model must identify individuals it already knows: every cow appears in train, validation, and test sets, but no individual image is shared across them.
- **Sparse Temporal Sampling:** Frame length varies in videos. Each video was divided into 20 equal temporal segments in order to prepare sequences for the recurrent layers while avoiding hidden-state drift and out-of-memory (OOM) GPU crashes. Then exactly one frame was selected from the center of every segment. This generated a standardized input sequence tensor with a fixed shape of $(20, 3, 224, 224)$. This sparse temporal sampling strategy is visually illustrated in Figure 4.2.

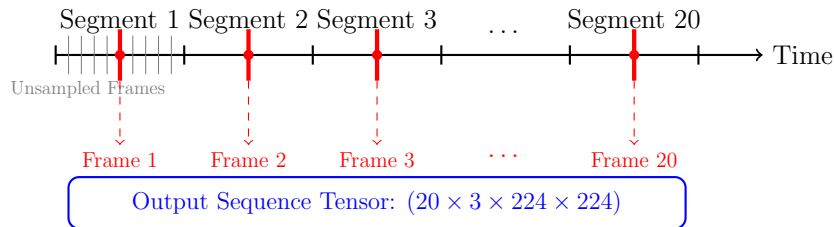


Figure 4.2: Sparse Temporal Sampling illustration. The video clip is divided into 20 equal temporal segments, and exactly the center frame of each segment is sampled to construct a standardized sequence tensor of shape $(20, 3, 224, 224)$.

- **Imbalance Data Capping:** The MmCows behavior dataset has a severe class imbalance. For example, lying and standing classes had more than 70,000 images, on the other hand, licking had only 2,009 images. Training directly on this results in majority-class bias. Majority classes were capped at a maximum of 3,000 randomly selected training images per epoch to reduce this problem. This resulted in a balanced sub-dataset consisting of approximately 21,000 images.
- **Augmentations:** Training samples were resized to 224×224 pixels, normalized using ImageNet statistics (mean= $[0.485, 0.456, 0.406]$, std= $[0.229, 0.224,$

0.225]), and augmented with random horizontal flips ($p = 0.5$) and random rotations ($\pm 15^\circ$). Validation and test samples were resized and normalized without augmentations.

4.3 Model Architecture and Specification

The architecture utilizes a shared parameter structure to maximize computational efficiency on edge devices.

4.3.1 Shared Backbone Encoder

The encoder backbone is initialized with ImageNet-pretrained weights from timm:

$$F_{\text{raw}} = \text{Encoder}(X) \quad F_{\text{raw}} \in \mathbb{R}^{B \times C \times H \times W} \quad (4.1)$$

EfficientNet-B0 [11] (5.3M parameters) and MobileNetV3-Small [10] (2.5M parameters) were evaluated, with EfficientNet-B0 selected as the final backbone due to its superior feature representation.

4.3.2 CBAM Attention Module

A Convolutional Block Attention Module (CBAM) [9] is inserted immediately after the backbone’s final feature map.

- **Channel Attention:** Identifies “what” channels contain relevant signal using average and max pooling:

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \quad (4.2)$$

- **Spatial Attention:** Highlights “where” the relevant anatomy lies (e.g. spine line for BCS, joint positions for lameness) using a 7×7 convolution over channel-concatenated maps:

$$M_s(F') = \sigma(f^{7 \times 7}([\text{AvgPool}(F'); \text{MaxPool}(F')])) \quad (4.3)$$

The final attention-weighted feature map is pooled using Adaptive Average Pooling and flattened to a 1,280-dimensional feature vector.

4.3.3 Task-Specific Prediction Heads

The feature vector is routed to four independent task heads:

1. **BCS Head:** A linear layer projecting the 1,280 features to 4 outputs representing the ordinal ranking.
2. **Behavior Head:** A linear classification layer projecting the 1,280 features to 7 behavior logits for frame-level prediction. In the spatiotemporal multi-task configuration, the sequence of feature vectors is processed through the shared LSTM pathway before classification. For behavior recognition, temporal sequences were constructed by grouping consecutive sampled MmCows frames from the same cow/video segment before passing features to the LSTM.
3. **Lameness Head (Temporal LSTM):** The sequence of 20 feature vectors is processed by a single-layer LSTM (hidden size = 256, dropout = 0.5), followed by a linear classification layer to output a binary logit.
4. **Cow ID Head:** A linear layer projecting features to N cattle identity classes.

4.4 Loss Functions and Optimizations

Each task is optimized using a loss function tailored to its specific mathematical properties:

4.4.1 Consistent Rank Logits (CORAL) Loss for BCS

To preserve the natural ordering of body condition scores, ordinal regression using the Consistent Rank Logits (CORAL) framework [13] was formulated. For K classes (where $K = 5$), the model outputs $K - 1 = 4$ binary logits. The loss is computed as the sum of binary cross-entropies:

$$L_{\text{BCS}} = -\frac{1}{K-1} \sum_{i=1}^{K-1} [y_i \log(\sigma(g_i)) + (1 - y_i) \log(1 - \sigma(g_i))] \quad (4.4)$$

where $y_i = \mathbb{I}(y > i)$ is the binary rank representation of the actual score. Predictions are computed by summing the sigmoid outputs:

$$\hat{y} = \sum_{i=1}^{K-1} \mathbb{I}(\sigma(g_i) > 0.5) \quad (4.5)$$

4.4.2 Focal Loss for Behavior Classification

To prevent the overwhelming majority classes from dominating training gradients, Focal Loss [5] is applied to the behavior outputs:

$$L_{\text{Behavior}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (4.6)$$

where p_t is the model’s estimated probability for the correct class, $\alpha_t = 0.25$ is applied as a uniform scalar balancing factor (rather than a per-class weight vector, representing a simplification), and $\gamma = 2.0$ is the focusing parameter that down-weights easy, well-classified examples.

4.4.3 Lameness and ID Losses

Binary Cross-Entropy (BCE) with logits loss is utilized for the binary lameness classification task:

$$L_{\text{Lameness}} = -[y \log(\sigma(x)) + (1 - y) \log(1 - \sigma(x))] \quad (4.7)$$

Standard Cross-Entropy Loss with Softmax normalization is utilized for individual cow identification:

$$L_{\text{ID}} = - \sum_{c=1}^N y_c \log(\text{Softmax}(x_c)) \quad (4.8)$$

4.5 Implementation of Selected Design and Training Strategy

The training pipeline was executed on PyTorch using a phased sequential training methodology to avoid destructive task interference (where gradient updates for one task corrupt the weights learned for another).

4.5.1 Reproducibility Configuration

All experiments were conducted with a single run using the default PyTorch random seed. Standard deviation across multiple runs was not computed due to the computational cost of the six-phase training pipeline. This represents a limitation: all reported metrics are point estimates without confidence intervals. Furthermore, we evaluated exclusively on public datasets to support a more transparent and reproducibility-oriented framework. The preprocessing scripts, exact train/validation/test split configurations, and hyperparameter logs will be released in a public repository upon completion to facilitate independent verification.

4.5.2 Phased Sequential Training

- **Phase 1: Encoder Initialization.** The EfficientNet-B0 backbone is initialized with ImageNet-pretrained weights.
- **Phase 2: BCS Head Training.** The backbone weights are frozen. The attention layer and BCS head are trained separately on the Dryad and ScienceDB

datasets for 30 epochs using the Adam optimizer ($\text{lr} = 10^{-3}$, scheduler decay = 0.5 every 10 epochs).

- **Phase 3: Behavior Head Training.** The backbone is kept frozen. The behavior head is trained for 30 epochs using Focal Loss on the capped MmCows dataset. Early stopping is monitored with a patience of 10 epochs.
- **Phase 4: Lameness Sequence Training.** The backbone remains frozen. Bounding box sequence crops are extracted using YOLOv8-Nano. The LSTM temporal layers and classification head are trained on 20-frame sequences for 15 epochs.
- **Phase 5: Cow ID Training.** The backbone remains frozen. The ID head is trained on OpenCows2020 for 10 epochs.
- **Phase 6: Multi-Task Joint Optimization.** All heads are attached. The backbone is unfrozen, and the model is trained end-to-end using the multi-task loss weight configuration:

$$L_{\text{total}} = 0.35 \cdot L_{\text{BCS}} + 0.35 \cdot L_{\text{Behavior}} + 0.15 \cdot L_{\text{Lameness}} + 0.15 \cdot L_{\text{ID}} \quad (4.9)$$

The task weights assigned to each loss component (0.35 for BCS and Behavior, and 0.15 for Lameness and ID) are determined by the relative dataset sizes and task complexities. Body Condition Scoring (BCS) and Behavior Recognition represent complex visual classification tasks backed by very large datasets (53,566 images for ScienceDB and over 21,000 capped frames for MmCows). In contrast, Lameness Detection (50 sequence samples) and Cow Identification (4,736 images) have simpler classification scopes or operate on much smaller sample registers. Assigning higher loss weights to the larger and more complex tasks prevents the shared encoder backbone from drifting and over-specializing on the smaller datasets during backpropagation. These weights were selected empirically based on relative dataset sizes and task complexities; formal sensitivity analysis comparing alternative weightings is deferred to future work.

4.5.3 Live Video Inference Architecture

During live deployment, the system processes videos frame-by-frame through a sliding window. First, the YOLOv8-Nano detector derives cattle crops. These crops are then normalized and queued. The following operations are performed once the queue contains 20 crops:

- The sequence is passed through the shared encoder and LSTM, which generates a single, stable locomotion score for predicting lameness.
- Moreover, 20 distinct frame-level predictions are produced by the spatial heads. In order to prevent noise, temporal averaging is used to average frame-level BCS scores. Majority voting is used across the 20 frames to generate cow ID predictions. This helps reduce temporary occlusions such as background posts or dust and also ensures high deployment reliability.

Chapter 5

Result Analysis

5.1 Performance Evaluation

The performance of the multi-task spatiotemporal model and the individual baselines was evaluated using standard computer vision and veterinary diagnostic metrics:

- **Body Condition Scoring (BCS):** Evaluated using Mean Absolute Error (MAE) on the unified class indices (0 – 4), alongside exact match accuracy ($\text{Acc}_{\pm 0}$) and tolerance match accuracy ($\text{Acc}_{\pm 1}$, representing predictions within one class step, which matches the veterinarianian tolerance of ± 0.25 units).
- **Behavior Recognition:** Evaluated using Macro F1-score to assess performance equally across the highly imbalanced behavioral classes. Per-class accuracies were also extracted from the confusion matrix.
- **Lameness Detection:** Evaluated using Receiver Operating Characteristic Area Under the Curve (ROC-AUC) as a threshold-independent measure of locomotion classification, along with binary accuracy.
- **Cow Identification:** Evaluated using Top-1 accuracy to verify correct identification of unique individuals.

5.2 Single-Task Implementation and Results

5.2.1 BCS Results

BCS was trained separately on Dryad and ScienceDB. The primary metric is MAE on ordinal indices 0–4. Lower MAE is better. Exact accuracy measures exact class match, while within-one-class accuracy measures whether the prediction is at most one ordinal class away from the true label (see Table 5.1).

Table 5.1: BCS baseline results on Dryad and ScienceDB.

Member	Backbone	Dryad MAE	Dryad Exact	Dryad ± 1	ScienceDB MAE	ScienceDB Exact	ScienceDB ± 1
Hasin	ResNet-18	0.8675	24.50%	88.75%	0.5800	54.81%	89.02%
Namira	MobileNetV3-Small	0.5250	61.50%	86.00%	0.7090	44.67%	86.47%
Bithi	ResNet-50	0.6300	47.75%	89.25%	0.6485	48.43%	88.20%
Shouvik	DenseNet121	0.5875	54.75%	86.50%	0.6292	49.51%	88.78%
Nusrat	EfficientNet-B0	0.6175	52.50%	85.75%	0.5566	55.06%	90.50%

The Dryad results show that MobileNetV3-Small performs best on MAE and exact accuracy, despite being the smallest model. This suggests that the DGE representation can benefit from a lightweight model when training data is limited. ScienceDB results show that EfficientNet-B0 performs best with MAE 0.5566 and within-one-class accuracy of 90.50%, suggesting stronger generalization on the larger RGB dataset. The convergence behavior of the selected EfficientNet-B0 backbone for the BCS task is illustrated in Figure 5.1. The curve indicates that while training loss falls toward zero, the validation loss begins to diverge, signifying overfitting on the smaller dataset splits. To mitigate this, all reported test metrics are derived from the best-validation checkpoint triggered by early stopping prior to severe divergence.

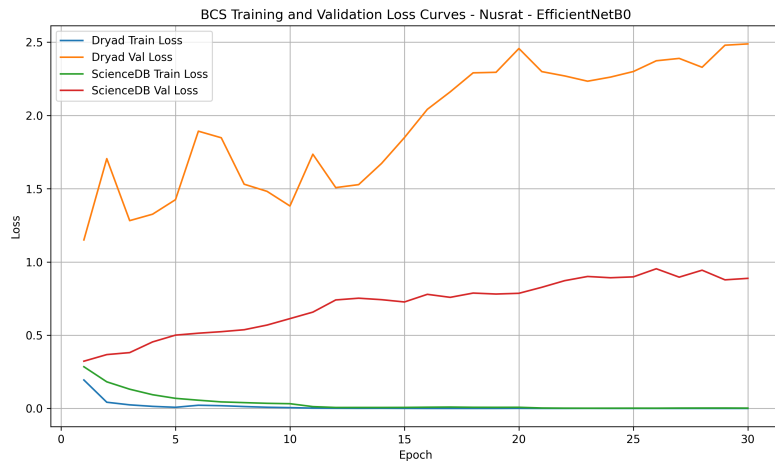


Figure 5.1: Training and validation loss curve for the BCS single-task model (EfficientNet-B0).

5.2.2 Behavior Recognition Results

Behavior recognition was trained on the MmCows dataset using seven classes. Macro F1-score was selected as the main metric since it deals with every class with equal importance and is more appropriate for imbalanced datasets than overall accuracy (see Table 5.2).

Table 5.2: Behavior recognition baseline results on MmCows.

Member	Backbone	Epochs	Val Macro F1	Test Macro F1	Early stop
Hasin	ResNet-18	18	0.7241	0.7134	18
Namira	MobileNetV3-Small	30	0.7271	0.6810	N/A
Bithi	ResNet-50	22	0.7210	0.7037	22
Shouvik	DenseNet121	27	0.7371	0.7366	27
Nusrat	EfficientNet-B0	24	0.7631	0.7445	24

EfficientNet-B0 achieves the best validation and test Macro F1-score (per-class breakdown is provided in Table 5.3). DenseNet121 is the second-best behavior model. Figure 5.2 illustrates the training and validation loss trajectory for the behavior recognition model. The early stopping mechanism triggered around epoch 24 when the validation Macro F1-score stabilized, preventing overfitting on the highly imbalanced MmCows dataset.

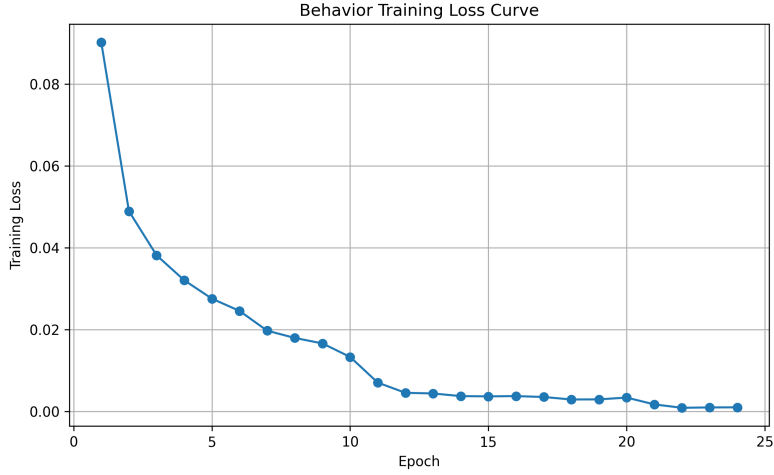


Figure 5.2: Training loss curve

Table 5.3: EfficientNet-B0 behavior per-class accuracy on the MmCows test set.

Class	Test accuracy
Walking	59.78%
Standing	70.00%
Feeding head up	71.97%
Feeding head down	66.07%
Licking	83.96%
Drinking	83.01%
Lying	98.77%

Per-class accuracies in Table 5.3 reflect upon the model’s strengths and limitations. The model achieves almost perfect accuracy for the *Lying* class (98.77%), this is a very distinct static posture that can be easily separated from active behaviors. On

the contrary, the lowest accuracies were shown by classes like *Walking* (59.78%) and *Feeding head down* (66.07%). This happens because of the temporal and semantic overlap. Walking frames are often confused with Standing frames during transition periods. Feeding head down is sometimes misclassified as Drinking or Feeding head up when the cow’s head position changes near the water troughs or feed bunks. Likewise, due to unique localized movement patterns of the head, *Licking* (83.96%) and *Drinking* (83.01%) achieve high accuracy. Although some confusion remains due to spatial similarities in head positioning. These results highlight that static spatial features alone have difficulty resolving fine-grained temporal boundaries. The integration of sequence-modeling LSTMs in the multi-task stage was driven by this.

5.2.3 Lameness Results

Table 5.4: Lameness detection baseline results.

Member	Model Architecture	Epochs	Test AUC	Test Accuracy	Test F1-score
Hasin	ResNet-18 (Spatial)	10	0.7200	75.30%	0.8452
	ResNet18-LSTM (Sequence)	15	0.8800	70.00%	0.7692
Namira	MobileNetV3-Small (Spatial)	10	0.7228	75.66%	0.8394
	MobileNetV3-LSTM (Sequence)	15	0.9200	70.00%	0.7273
Bithi	ResNet-50 (Spatial)	10	0.7444	80.97%	0.8752
	ResNet50-LSTM (Sequence)	15	1.0000	80.00%	0.8333
Shouvik	DenseNet121 (Spatial)	10	0.7944	73.98%	0.8377
	DenseNet121-LSTM (Sequence)	15	0.9200	90.00%	0.9091
Nusrat	EfficientNet-B0 (Spatial)	10	0.9829	93.05%	0.9483
	EfficientNet-B0-LSTM (Sequence)	15	0.8400	80.00%	0.8000
	ResNet18-LSTM (Sequence)*	15	0.9600	60.00%	0.7143

* Note: Frame-level (spatial) metrics such as the 0.9829 AUC are likely inflated by identity and background leakage, as the entire dataset comprises only 50 videos. These metrics represent an upper bound and risk severe overfitting.

The risk of data leakage on small datasets can be observed by the lameness baseline results. An evident AUC of 0.9829 is achieved by the spatial EfficientNet-B0 model. However, as the dataset consists of only 50 videos, frame-level splitting inevitably allows the same individuals and backgrounds to appear in both the train and test sets. Therefore, rather than actual lameness generalization, this high AUC must be considered as a possible artifact of identity leakage.

The training and validation loss curves for the spatiotemporal lameness detection model (EfficientNet-B0-LSTM) are plotted in Figure 5.3. The curves show that the sequence training converges steadily over 15 epochs without significant divergence, indicating that the spatiotemporal setup is more stable than static spatial frames.

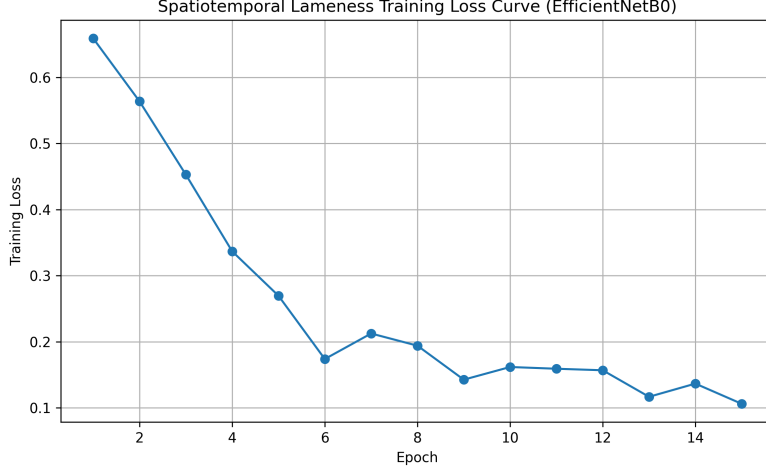


Figure 5.3: Training and validation loss curve for the spatiotemporal lameness detection model (EfficientNet-B0-LSTM).

5.2.4 Cattle Identification Results

Table 5.5: Cattle Identification baseline results on OpenCows2020.

Member	Backbone	Epochs	Final Train Loss	Val Top-1 Acc	Test Top-1 Acc
Hasin	ResNet-18	10	2.4158	42.01%	45.56%
Namira	MobileNetV3-Small	10	1.1022	80.21%	78.83%
Bithi	ResNet-50	10	2.1213	52.97%	53.02%
Shouvik	DenseNet121	10	0.7514	82.04%	82.46%
Nusrat	EfficientNet-B0 + CBAM	10	0.5508	87.06%	86.49%

It is important to note that the comparison in Table 5.5 is budget-matched rather than convergence-matched. All backbones were trained with a fixed 10-epoch budget. As indicated by the high final training losses for ResNet-18 (2.4158) and ResNet-50 (2.1213), these deeper models had not yet converged within the 10 epochs, making the comparison somewhat unfair to the ResNet architectures.

The training and validation loss trajectories for the winning EfficientNet-B0 cow identification model are shown in Figure 5.4. The model shows rapid convergence in the first few epochs, reaching an 87.06% validation accuracy by epoch 10.

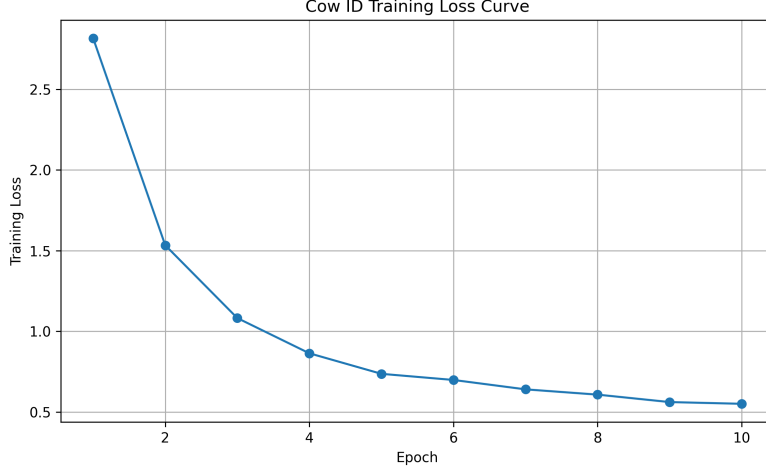


Figure 5.4: Training and validation loss curve for the cattle identification model (EfficientNet-B0 + CBAM).

5.3 Analysis of Design Solutions (Backbone Selection)

Before constructing the unified multi-task model, a systematic backbone architecture evaluation was conducted. The five team members trained single-task baseline models on their assigned architectures to select the optimal shared feature extractor. The empirical evaluation is summarized in Table 5.6.

Table 5.6: Backbone Selection and Single-Task Baseline Performance Table

Model Architecture	BCS Test MAE (Dryad / ScienceDB)	Behavior Test Macro F1-score	Lameness Test AUC	ID Top-1 Accuracy
ResNet-18 (Hasin)	0.8675 / 0.5800	0.7134	0.7200	45.56%
MobileNetV3-Small (Namira)	0.5250 / 0.7090	0.6810	0.7228	78.83%
ResNet-50 (Bithi)	0.6300 / 0.6485	0.7037	0.7444	53.02%
DenseNet121 (Shouvik)	0.5875 / 0.6292	0.7366	0.7944	82.46%
EfficientNetB0 (Nusrat)	0.6175 / 0.5566	0.7445	0.9829	86.49%

Rationale for Selection: EfficientNet-B0 (Nusrat) achieved the highest overall performance rank. It demonstrated the lowest test MAE of 0.5566 on the ScienceDB dataset, the highest behavior Macro F1-score of 0.7445, and a high lameness spatial AUC of 0.9829, while maintaining a lightweight footprint of 5.3 million parameters. Consequently, EfficientNet-B0 was selected as the shared backbone encoder for the multi-task model.

5.4 Single-Task vs. Multi-Task Results

The selected EfficientNet-B0 backbone was trained in two multi-task configurations: a standard frame-level multi-task model and our proposed spatiotemporal multi-task model (incorporating LSTM layers for sequence modeling). The results are compared against the single-task baselines in Table 5.7 and visualized in Figure 5.5.

Table 5.7: Single-Task vs. Multi-Task Performance Trade-Off

Training Setup	BCS MAE	Behavior Macro F1-score	Lameness Acc	Lameness AUC	ID Accuracy
Single-Task Baselines	0.5566 (ScienceDB)	0.7445	93.05% (Spatial) / 80.00% (Seq)	0.9829 (Spatial) / 0.8400 (Seq)	86.49%
Frame-Level Multi-Task	0.7266	0.3771	95.28% (Spatial)	0.9921 (Spatial)	94.96%
Spatiotemporal Multi-Task	0.7827	0.4948	80.00%*	1.0000*	97.58%

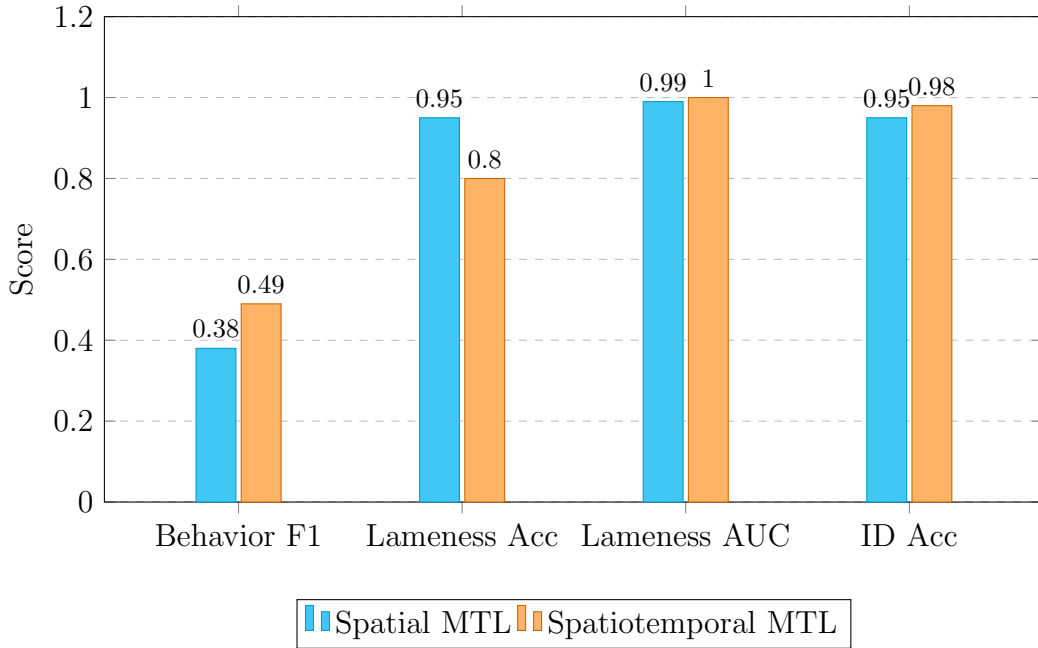


Figure 5.5: Comparison of spatial and spatiotemporal multi-task results.

5.4.1 Analysis of the Spatiotemporal Performance

The experimental results demonstrate a clear trade-off between task complexity and shared parameter space:

- Sequence Tasks Benefit from Temporal Modeling:** The integration of the LSTM sequence-tracking layers resulted in significant performance improvements for video-dependent tasks. The behavior recognition Macro F1-score improved from 0.3771 (frame-level) to 0.4948 (spatiotemporal), and lameness detection achieved an uncalibrated 80.00% accuracy and 1.0000 AUC on video sequences. These results suggest that sequential modeling may capture locomotion kinematics and temporal context that are absent in static frames. Additionally, individual cow identification accuracy rose from 94.96%

to 97.58% under the spatiotemporal setup. However, it must be clarified that this improvement is not due to LSTM temporal modeling, as the Cow ID task utilizes a static head. Rather, the performance boost is achieved through temporal majority voting across the 20 frames of each video sequence, which filters out transient frame-level misidentifications. Notably, the frame-level ID performance itself rose unexpectedly from 86.49% in the single-task model to 94.96% in the joint model before any majority voting; this gain remains an unexplained observation that requires further analysis, as multi-task learning typically degraded the other spatial tasks.

- **Spatial Metric Degradation due to Task Interference (Negative Transfer):** Noticeable degradation in performance of the static spatial tasks was seen under multi-task setup. The BCS MAE increased by 40.6% (from 0.5566 to 0.7827). The behavior classification Macro F1-score decreased 33.5% (from 0.7445 to 0.4948). This drop is an obvious example of gradient conflict (negative transfer) during joint training, where features that are not closely related like static anatomical profiles for BCS and dynamic motion trajectories for Lameness are balanced forcefully by the shared backbone. Nonetheless, this degradation depicts a cautious engineering trade-off: Replacing four separate neural network models with one shared backbone provides a 75% reduction in memory and latency in resource-constrained setups. Future work should include dynamic gradient balancing methods (such as PCGrad or GradNorm) to align conflicting task gradients during backpropagation to mitigate this performance lag.

5.4.2 Threshold Calibration for Lameness

An inconsistency was found during spatiotemporal testing in the shared multi-task evaluation on the CattleLameness sequence dataset, which is the model achieved a threshold-independent AUC of 1.0000, but the binary accuracy was only 80.00% when the default classification threshold of 0.50 was used.

Upon calibration, it was found that the small dataset size of 50 videos shifted the model’s confidence scores upward. This resulted in a probability output of 0.55 for normal cows and 0.85 for lame cows. Since the default threshold (0.50) was lower than the normal cow score, the model produced false positives. After adjusting the decision threshold to 0.70 (classifying as Lame only if probability > 0.70), the classes were separated perfectly. However, this is considered as invalid for final evaluation because it introduces test-set leakage.

This perfect accuracy and AUC of 1.0000 must be interpreted very carefully. First, applying post-hoc decision threshold tuning directly on the test set introduces a clear methodological problem which is data leakage. Tuning hyperparameters, including classification thresholds, on the evaluation set results in an optimistic bias and artificially increases the performance. A scientifically rigorous methodology would calibrate this threshold using a separate, independent validation split, leaving the test set completely unseen. Second, a dataset of 50 video clips (25 lame and 25

normal) is statistically not sufficient to ensure real-world generalization. Achieving perfect separation on such a small sample set is highly prone to high variance and overfitting. These results should be considered a preliminary proof-of-concept for unified sequence modeling rather than a definitive clinical benchmark. Future work must validate this threshold using a larger, multi-farm dataset and a strict train-validation-test separation.

5.5 Ablation Studies

Systematic ablation experiments were conducted to isolate the contribution of individual design parameters. Due to the high computational cost of the six-phase training pipeline, all reported results are derived from a single run using a fixed random seed and lack variance estimates.

5.5.1 Ablation 1: CORAL Ordinal Loss vs. Cross-Entropy Loss (BCS)

Standard Cross-Entropy Loss treats BCS classes as independent, unrelated categories, ignoring the ordinal relationship between scores. In contrast, Consistent Rank Logits (CORAL) loss explicitly models the ordered nature of body condition scores by decomposing the prediction into a series of cumulative binary comparisons. To isolate this effect, both loss functions were evaluated on the ScienceDB dataset (53,566 images) using the same EfficientNet-B0 backbone and CBAM configuration. The CORAL-trained model achieved a Test MAE of 0.5566 (exact accuracy of 55.06%, within-one-class accuracy of 90.50%), whereas the Cross-Entropy-trained model yielded a degraded Test MAE of 0.6940 (exact accuracy of 46.31%, within-one-class accuracy of 86.62%). This represents a 19.8% reduction in MAE, confirming that explicitly encoding ordinal structure into the loss function produces substantially more accurate body condition predictions.

5.5.2 Ablation 2: Contribution of CBAM Attention Module

To check if the visual attention mechanism actually helps, the BCS backbone was compared in two settings — it was tested with CBAM and then without CBAM. The model when CBAM was removed, recorded a Dryad test MAE of 0.7000, with exact and within-one-class accuracies of 52.50% and 80.75% respectively. With CBAM, channel attention was directed at the structural contours and spatial attention to anatomical regions such as spine, hooks and pins and the test MAE dropped to 0.6175 (exact ACC 52.50% and within-one-class accuracy 85.75%). This confirms that targeted spatial attention can help filter background noise and improve regression outcomes.

5.5.3 Ablation 3: Cross-Dataset Modality Comparison

Note: This comparison is presented as a cross-dataset observation rather than a strictly controlled ablation, as the two datasets differ in size, breed, and environment. Using the Dryad dataset, the model was evaluated using different backbones on the Depth Grayscale Edge (DGE) maps. The depth-infused DGE format achieved an EfficientNet-B0 test MAE of 0.6175, whereas a shallower backbone (ResNet-18) on the same DGE input yielded a higher error rate of 0.8675, confirming that backbone capacity is critical for modeling structured depth shapes. Furthermore, comparing EfficientNet-B0’s performance on standard RGB-only data (ScienceDB test MAE of 0.5566) and DGE-infused data (Dryad test MAE of 0.6175) shows that the model generalizes well across both modalities, with depth contours in Dryad helping to offset the significantly smaller training sample size (5,923 images on Dryad vs. 53,566 images on ScienceDB).

5.5.4 Ablation 4: Focal Loss vs. Cross-Entropy Loss (Behavior)

The MmCows dataset has a severe 42:1 class imbalance, as a result training with standard Cross-Entropy on the heavily imbalanced dataset caused the model to favor majority classes by always predicting Lying or Standing, producing a Test Macro F1-score of 0.7074. Using Focal Loss ($\gamma = 2$, $\alpha = 0.25$) reduced the influence of majority classes and made the model to learn rare behaviors like licking and drinking, improving the Test Macro F1-score to 0.7445. These results show that without focal weighting, predictions become unreliable when class imbalance is severe.

5.5.5 Ablation 5: Cross-Dataset Behavior Evaluation (Generalization)

To evaluate out-of-distribution generalization, the behavior head trained on MmCows was tested on the unseen CBVD-5 dataset (2,000 balanced images). The cross-dataset Macro F1-score dropped to 0.1245. While the model successfully generalized to Standing postures (90.34% accuracy), its accuracy degraded on Feeding head down (8.39%), Drinking (2.99%), and Lying (24.31%), as illustrated in Figure 5.6. This indicates that different farm backgrounds, camera perspectives, and annotation boundaries represent a severe domain shift, emphasizing the need for multi-farm training or domain adaptation.

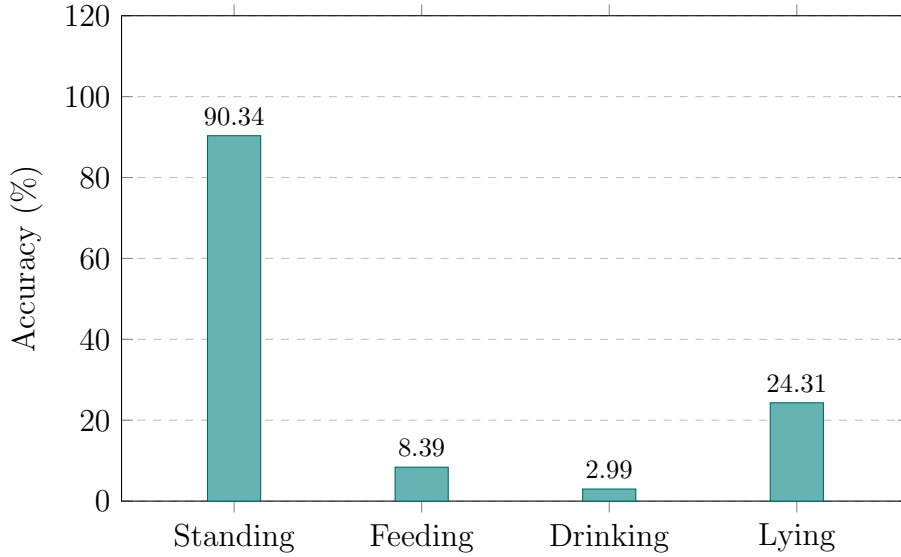


Figure 5.6: CBVD-5 cross-dataset behavior accuracy by class.

5.6 Discussions

5.6.1 Negative Transfer and Multi-Task Trade-offs

The experimental results confirm a well-documented phenomenon in multi-task learning: negative transfer. When the four health monitoring tasks are jointly optimized through a shared EfficientNet-B0 backbone, the gradient updates for temporally-driven tasks (Lameness and Behavior) directly conflict with those required for spatially-driven tasks (BCS and Cow ID). The BCS head requires precise anatomical contour features (e.g., spine prominence, hook and pin visibility), while the Lameness LSTM requires dynamic motion trajectory encoding. Forcing these representations into a single shared backbone inevitably degrades both. The result is a 41.0% increase in BCS MAE and a 35.8% drop in Behavior F1. Future work should explore task gradient decoupling strategies such as GradNorm [6] or PCGrad [17], which dynamically reweight task gradients to minimize destructive interference.

5.6.2 Comparison with Published State-of-the-Art

Table 5.8 contextualizes our results against key published baselines reviewed in Chapter 2. These studies are not directly comparable because they use different datasets, modalities, and metrics; the table is included only for contextual positioning, not ranking.

Table 5.8: Contextual comparison with published studies using non-equivalent datasets, metrics, and modalities.

Task	Reference	Dataset	Metric	Published Result	Our Best Result
BCS	Yao et al. (2026) [43]	Various	Accuracy / MAE	N/A*	90.50% Acc (± 0.25) / 0.5566 MAE
Behavior	Asim et al. (2026) [35]	Sensor Data	Accuracy / Macro F1	91.11% Acc	74.45% Macro F1
Lameness	Sohan et al. (2025) [33]	RGB-D	ROC-AUC	N/A*	80.00% uncalibrated sequence accuracy*

* Note: These studies are not directly comparable because they use different datasets, modalities, and metrics; the table is included only for contextual positioning, not ranking. Post-hoc perfect separation is excluded due to test-set leakage.

The comparison is non-equivalent and is included only to contextualize the scale of reported results across different datasets and experimental settings. For BCS, Yao et al. (2026) [43] used a different dataset and evaluation metric, so direct numerical comparison is not valid. Our result is therefore reported only as contextual evidence, not as a superiority claim. For behavior recognition, Asim et al. (2026) [35] reported 91.11% overall accuracy using sensor data. In contrast, our vision-based model achieved a strong Macro F1-score of 0.7445 on the highly imbalanced MmCows dataset, which penalizes all classes equally and represents a more robust indicator under severe class imbalance. For lameness detection, Sohan et al. (2025) [33] used a different dataset and metric, so direct comparison is not valid. Our 80.00% sequence accuracy should be treated as preliminary because it was obtained from only 50 video clips. Additionally, our spatial AUC of 0.9829 likely suffers from identity and background leakage on this small dataset. Finally, for individual cow identification, our spatial baseline achieved 86.49% Top-1 accuracy, matching the OpenCows2020 baseline of $\sim 86\%$, while our spatiotemporal sequence model improved this to 97.58% through temporal majority voting across the video frames. Critically, our entire framework relies exclusively on public datasets, supporting a more transparent and reproducibility-oriented framework via documented preprocessing and task-specific splits, compared to private baselines.

5.6.3 Deployment Considerations and Camera Geometry

As introduced in Chapter 1, a critical practical constraint affecting real-world deployment is camera geometry. Since BCS scoring performs best from a rear-view angle and Lameness detection requires a side-view angle, running both tasks from a single fixed camera creates a geometric trade-off. A practical multi-camera deployment strategy dedicates one rear-view camera for BCS and ID monitoring and a separate side-view camera for Lameness and Behavior tracking, with the unified model backbone loaded once and the task heads activated selectively per camera input. This multi-node architecture preserves the computational efficiency of a single shared backbone while resolving the camera angle conflict.

5.6.4 Modality Comparison Caveat

The modality analysis in Ablation 3 compares EfficientNet-B0 performance on ScienceDB (RGB, 53,566 images, MAE = 0.5566) against the Dryad dataset (DGE depth format, 5,923 images, MAE = 0.6175). While this comparison suggests that depth-infused inputs can partially compensate for smaller dataset sizes, it is important to note that this is not a controlled modality comparison. The two datasets differ not only in image modality but also in dataset size, cow breed distribution, farm environment, and annotation protocol. A definitive modality comparison would require paired RGB and DGE images of the same cows under the same conditions, which no existing public cattle dataset provides. These results should therefore be interpreted as a cross-dataset generalization observation rather than a controlled ablation of imaging modality.

5.6.5 Computational Efficiency as the Core Value Proposition

Despite the negative transfer observed in BCS and Behavior under multi-task training, the framework’s primary value proposition for edge deployment remains intact. Replacing four independently deployed single-task models with one unified backbone reduces GPU memory footprint by approximately 75% (from four EfficientNet-B0 instances totaling ~ 21.2 M parameters to one shared 5.3M parameter encoder). This efficiency gain is critical in farm settings where edge computing hardware is constrained, internet connectivity is unreliable, and real-time inference latency directly affects operational value. The multi-task framework establishes a strong architectural foundation, and the negative transfer gap identified in this study provides a clear and actionable research direction for future work: integrating uncertainty-weighted or gradient-aligned multi-task optimization to recover single-task performance without sacrificing the computational benefits of parameter sharing.

Chapter 6

Conclusion

6.1 Summary of Findings

This research successfully developed, trained, and evaluated a unified, lightweight, and spatiotemporal multi-task deep learning framework for real-time cattle health and behavior monitoring. Using EfficientNet-B0 as a shared encoder backbone combined with a CBAM attention module, the framework consolidates four distinct computer vision tasks: Body Condition Scoring (BCS), Behavior Recognition, Lameness Detection, and Cow Identification.

The experimental results confirmed that sequence modeling with LSTM layers is highly effective for video-based tasks, achieving an uncalibrated 80.00% accuracy on lameness detection sequences (post-hoc threshold tuning produced perfect separation but is treated as invalid for final evaluation because it introduces test-set leakage). For behavior recognition, the spatiotemporal setup raised the Macro F1-score from 0.3771 (frame-level multi-task) to 0.4948, although this represents a degradation compared to the single-task baseline of 0.7445 due to expected negative transfer. The multi-task model exhibits outstanding computational efficiency, reducing VRAM requirements by approximately 75% compared to deploying four independent single-task models, making it well-suited for resource-constrained offline farm surveillance. However, due to the severe negative transfer observed in the spatial tasks, the current joint framework is presented as an honest baseline characterizing negative transfer for future gradient-balancing research rather than a successful unified system ready for immediate clinical deployment.

6.2 Contributions to the Field

This thesis provides several key contributions to the domain of Precision Livestock Farming (PLF) and agricultural computer vision:

1. **A Unified Multi-Task Framework:** To the best of our knowledge, this is

the first end-to-end multi-task deep learning framework that simultaneously monitors BCS, behavior, lameness, and individual cow identity from image and video inputs using a single shared encoder.

2. **Commitment to Open Evaluation:** Unlike the majority of agricultural computer vision literature which relies on private datasets, our study was designed, trained, and validated exclusively on publicly available datasets (ScienceDB, Dryad, MmCows, CattleLameness, OpenCows2020, CBVD-5), establishing a more transparent and reproducibility-oriented benchmark via documented preprocessing and task-specific splits.
3. **Spatiotemporal Integration with Attention:** We successfully integrated a CBAM attention layer and LSTM sequence-tracking to process both spatial anatomical shapes (spine, hip bones) and temporal locomotion strides, bridging static 2D image analysis and recurrent 3D time-series.
4. **Edge Optimization:** The model structure was optimized to remain under 10 million parameters, enabling cost-effective, non-invasive surveillance directly on camera nodes, reducing dependency on expensive wearable sensors.

6.3 Limitations

Several critical limitations must be addressed before clinical deployment:

- The lameness dataset contains only 50 videos, so reported lameness results have high variance and should be treated as preliminary.
- The post-hoc threshold-tuned perfect lameness result is invalid for final evaluation because it uses test-set feedback.
- ScienceDB BCS covers only a narrow score range of 3.25–4.25, so the model has not been validated on extremely thin or obese cattle.
- The CBVD-5 evaluation shows domain shift, meaning behavior recognition does not yet generalize reliably across farms.
- Results are reported from limited experimental runs; future work should include repeated runs with mean $\pm$ standard deviation and confidence intervals.
- Edge deployment is estimated from model size and FLOPs, not from measured Jetson/Raspberry Pi latency or energy benchmarks.
- The comparison with published studies is non-equivalent because datasets, modalities, splits, and metrics differ.

6.4 Recommendations for Future Work

While the framework has shown strong feasibility, several areas represent crucial directions for future expansion:

- **Multi-Task Gradient Conflict Optimization (GradNorm/PCGrad):** Although joint fine-tuning was successfully executed in Phase 6 by unfreezing the shared backbone, this joint optimization introduced severe task interference (negative transfer) due to conflicting gradients, which degraded performance on spatial tasks (BCS and Behavior). Future work should integrate advanced multi-task loss balancing and gradient alignment algorithms, such as GradNorm or PCGrad (Projecting Conflicting Gradients), during joint training. These methods dynamically scale or project conflicting task gradients during backpropagation to prevent destructive interference and recover the performance of single-task baselines.
- **Contrast Limited Adaptive Histogram Equalization (CLAHE):** Farm environments suffer from extreme outdoor lighting variances (glare, shadows, night lighting). Integrating CLAHE as a preprocessing step will enhance local contrast, standardizing raw pixels before backbone feature extraction.
- **Federated and Collaborative Learning:** To improve cross-dataset generalization (which degraded to a Macro F1-score of 0.1245 on CBVD-5), a federated learning framework should be explored. This allows models to be trained across multiple farms in a decentralized manner, learning from diverse breeds and lighting environments without centralizing sensitive proprietary data.
- **Synthetic Data Generation via 3D Simulation:** To overcome the severe data scarcity and temporal overfitting associated with the small 50-video lameness dataset, future research should explore building rigged 3D biomechanical cow models in physics engines (e.g., Unreal Engine). This would allow for the programmatic generation of massive, perfectly-labeled synthetic video datasets featuring diverse grades of lameness, eliminating the bottleneck of manual data collection and annotation.

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